

In[27]:= (*Defining the Coordinates*)

```
coords = {t, r,  $\theta$ ,  $\phi$ };
```

(*Defining the Metric*)

```
g = {{-Exp[2  $\Phi$ [r]], 0, 0, 0}, {0, Exp[2  $\nu$ [r]], 0, 0}, {0, 0,  $r^2$ , 0}, {0, 0, 0,  $r^2 \sin[\theta]^2$ }};
```

```
ginv = Simplify[Inverse[g]];
```

(*Defining the Christoffel symbols*)

```
 $\Gamma[\rho\_ , \mu\_ , \omega\_ ] := 1/2 \text{Sum}[ginv[[\rho, \sigma]] (D[g[[\sigma, \omega]], coords[[\mu]] +$   

 $D[g[[\sigma, \mu]], coords[[\omega]] - D[g[[\mu, \omega]], coords[[\sigma]]]), \{\sigma, 1, 4\}];$ 
```

(*Defining the Ricci tensor*)

```
R[\mu\_ , \omega\_ ] := Sum[D[\Gamma[\rho, \omega, \mu], coords[[\rho]] - D[\Gamma[\rho, \rho, \mu], coords[[\omega]]] + Sum[  

 $\Gamma[\rho, \rho, \lambda]$   $\Gamma[\lambda, \omega, \mu] - \Gamma[\rho, \omega, \lambda]$   $\Gamma[\lambda, \rho, \mu]$ , {\lambda, 1, 4}], {\rho, 1, 4}];
```

(*Defining the Ricci scalar*)

```
Rscal = Sum[ginv[[\mu, \omega]] R[\mu, \omega], {\mu, 1, 4}, {\omega, 1, 4}];
```

(*Calculating Einstein tensor*)

```
G[\mu\_ , \omega\_ ] := Simplify[R[\mu, \omega] - 1/2 g[[\mu, \omega]] Rscal];
```

(*Printing non-zero elements of Einstein tensor*)

```
Print["Non-zero components of Einstein tensor G_{\mu\nu}:"];
```

```
Do[Module[{val = Simplify[G[\mu, \omega]],
```

```
  If[val != 0, Print["G_", coords[[\mu], coords[[\omega], " = ", val]]], {\mu, 1, 4}, {\omega, 1, 4}];
```

Non-zero components of Einstein tensor G_{\mu\nu}:

$$G_{tt} = \frac{e^{-2\nu[r]+2\Phi[r]}(-1 + e^{2\nu[r]} + 2r\nu'[r])}{r^2}$$

$$G_{rr} = \frac{1 - e^{2\nu[r]} + 2r\Phi'[r]}{r^2}$$

$$G_{\theta\theta} = e^{-2\nu[r]} r (\Phi'[r] + r\Phi'[r]^2 - \nu'[r] (1 + r\Phi'[r]) + r\Phi''[r])$$

$$G_{\phi\phi} = e^{-2\nu[r]} r \sin[\theta]^2 (\Phi'[r] + r\Phi'[r]^2 - \nu'[r] (1 + r\Phi'[r]) + r\Phi''[r])$$